

Education

Washington University in St. Louis

St. Louis, MO

Ph.D. in Computer Science

Sept. 2025 – Present

Research area: Trustworthy AI-enabled cyber-physical systems and secure agentic AI systems

Columbia University

New York, NY

B.S. in Computer Science, GPA: 3.89/4.0

Sept. 2023 – Jun. 2025

Pitzer College

Claremont, CA

B.A. in Management Engineering, GPA: 3.8/4.0

Sept. 2020 – May 2023

Research Interests

Trustworthy AI-enabled cyber-physical systems; physical-world adversarial robustness; secure agentic AI systems; Internet of Agents; robotics and perception security.

Research Experience

Washington University in St. Louis, Computer Security and Privacy Laboratory

St. Louis, MO

Ph.D. Student Researcher | Advisor: Prof. Ning Zhang

Sept. 2025 – Present

Research Area: Trustworthy AI-enabled cyber-physical and agentic systems**Project: Physical-World Robustness of Object Detectors**

- Studied physical-world adversarial patch robustness against object detectors, focusing on how attacks survive the physical-to-digital pipeline from actuation and environmental transmission to sensing and preprocessing.
- Built and used simulation and physical testbeds involving YOLOv5, CARLA, Gazebo, ROS2, PX4/drone setups, real camera capture, and printed adversarial objects/patches.
- Conducted a literature survey of approximately 130 papers on physical adversarial examples and organized prior work by attack medium, victim task, and pipeline-stage coverage.
- Evaluated sensing and preprocessing factors affecting adversarial-example reliability, including focus, field of view/focal length, camera/ISP variation, JPEG compression, resizing/interpolation, illumination, viewpoint, and distance.

Project: Internet of Agents

- Developed an initial lab-scale Internet of Agents prototype for LLM agents, supporting agent publication and discovery through an ADS server based on Agntcy A2A/ADS concepts.
- Exploring security challenges in open agent-to-agent ecosystems, including agent certification, privacy exposure, capability attestation, reputation, and trust establishment.

Southeast University

Nanjing, China

Research Assistant | Mentor: Prof. Fang Dong

May 2024 – Aug. 2024

Project: Sub-Second QoS-Aware Scheduling for Dynamic Serverless Workloads

- Designed a QoS-aware scheduling system for dynamic serverless workloads, targeting cold-start mitigation and tail-latency reduction under sub-second scheduling constraints.
- Integrated process-level checkpoint/restore with runtime scheduling to reduce function initialization overhead and improve responsiveness.
- Deployed the prototype on Kubernetes and evaluated scheduling behavior under variable workload and resource conditions.
- Analyzed tradeoffs among latency, resource utilization, and QoS compliance for time-sensitive serverless applications.

Research Software and Infrastructure

Internet of Agents Prototype

2026

- Developed a lab prototype for agent publication and discovery using an ADS server; used for early research exploration of security and trust issues in agent-to-agent ecosystems.

Gazebo Adversarial Example Testing Environment

2025 – 2026

- Developed a Gazebo-based testing environment for evaluating physical adversarial examples in simulated robotic and cyber-physical settings.

Electromagnetic Interference Ledger & Registry (Emily)

2026

- Developed an internal-testing prototype for the Caltech Department of Astronomy to support electromagnetic-interference record management and registry workflows.

Selected Research and Systems Projects

Exploratory Evaluation of Vision-Language-Action Models on LIBERO

2026

- Evaluated instruction-following limitations of VLA models on LIBERO through prompt ablations and scene-manipulation tests, finding that benchmark success can be driven by scene-layout priors rather than robust language grounding.

S-Python: Strongly Typed Python-like Language

Jan. 2024 – May 2024

- Built a strongly typed Python-like language as a team project, contributing across parsing, semantic checking, object-oriented features, garbage collection, anonymous functions, and testing.

Harvey Mudd College

Claremont, CA

Research Student | Mentor: Prof. Geoff Kuenning

May 2022 – Jul. 2022

Project: File-System Model Checking

- Contributed to a SPIN-based model-checking project for file-system behavior; tested workloads involving ext4, jffs2, btrfs, xfs, and betrfs, and documented debugging workflows for reproducible experiments.

Teaching Experience

Columbia University, Department of Computer Science

New York, NY

Teaching Assistant, COMS W4115 Programming Languages and Translators

Sept. 2024 – Dec. 2024

- Supported a 40-student upper-level programming languages and compilers course; held office hours and graded programming assignments, exams, and final language-implementation projects.

Harvey Mudd College, Department of Computer Science

Claremont, CA

Head Teaching Assistant, CS 70: Data Structures/Program Development

Sept. 2022 – Dec. 2022

- Served as senior TA; held three weekly office hours, evaluated homework and exams, and assisted other TAs with course-support questions.

Harvey Mudd College, Department of Computer Science

Claremont, CA

Teaching Assistant, CS 5: Introduction to Computer Science

Feb. 2022 – May 2022

- Supported lab sessions, weekly student meetings, and homework help for introductory programming and theoretical computer science topics.

Professional Service

- External reviewer, IEEE Transactions on Information Forensics and Security, 2026.

Technical Skills

- **Programming:** Python, C/C++, Java, R, Bash, OCaml, MATLAB
- **Systems and Infrastructure:** Linux, Docker, Kubernetes, ROS2, PX4, Git, L^AT_EX
- **ML / Robotics / Simulation:** PyTorch, TensorFlow, YOLOv5, CARLA, Gazebo, OpenCV, LIBERO
- **Agentic AI:** LLM agents, agent discovery, A2A/ADS-based agent infrastructure
- **Languages:** Mandarin Chinese, English, Japanese